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A superconvergence result for discontinuous Galerkin methods applied to elliptic problems

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Abstract

This paper presents a theoretical and numerical study of a class of discontinuous Galerkin methods that shows the approximation of the gradient superconverges at the zeros of the Legendre polynomials on a model 1D elliptic problem. Numerical experiments validate the theoretical results.
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1. Introduction

It is well known that for two-point boundary problems we can identify special points that converge at a faster rate than that predicted by global a priori error estimates. The rapid convergence of these points can be used in the design of post-processing techniques to enhance the accuracy of computations or in the design of error indicators for h-refinement. This superconvergence behavior of the classical finite element method has been analyzed for several years. In [12], Douglas and Dupont showed that for a general class of two-point boundary problems, the rate of convergence at the mesh points is of order $O(h^{2p})$ when polynomials of degree p are used, and that for a model problem the finite element approximation coincides with the exact solution. In [4], Bakker proved that the gradient is superconvergent of $O(h^{p+1})$ at the zeros of a Legendre polynomial; in [5], he showed that the finite element solution converges with an order $O(h^{p+2})$ at the Lobatto points. In both cases, the rates are one order better that the expected global order of convergence. An extensive compilation of superconvergence results for the classical C^0 finite element method can be found in the monograph by Wahlbin [15].

In recent years there has been much interest in the application of the so-called discontinuous Galerkin (DG) methods to elliptic problems. Originally designed for conservation laws, these methods are locally conservative and are able to capture steep fronts. DG methods may be regarded as high-order extensions of the classical finite volume method. Since no inter-element continuity is imposed, these methods can be

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defined on very general meshes, including non-conforming meshes. Moreover, polynomials of arbitrary degree can be used at each element, making these methods suitable for *hp*-refinement.

The analysis of superconvergent points for DG methods has been limited to hyperbolic problems. In [1], Adjerid et al. showed that for the model problem $u' - f(u) = 0$ the finite element solution superconverges at the zeros of Radau polynomials. This result has been successfully used in the design of efficient error indicators [1,7]. On the other hand, the existence of such points for elliptic problems has barely been addressed in the literature. In [13], Celani and Dutra do Carmo investigated the possibility of applying the patch recovery technique of Zienkiewicz and Zhu [16] to the DG method proposed by Baumann and Oden [6]. In this technique a more accurate approximation of the gradient is obtained by a least-square fitting using a suitable set of sampling points. Typically, such a set consists of points where the computed gradient is known to superconverge (e.g., the zeros of Legendre polynomials [4]). This approach was followed in [13]. However, it is not known if the gradient superconverges at these points in DG methods.

This paper investigates the existence of superconvergent points for DG methods applied to elliptic problems. We extend some well-known results of superconvergence to a class of DG methods. In particular, we prove that under certain conditions a special gradient operator superconverges at the zeros of a Legendre polynomial. More precisely, we use a gradient operator that can be obtained naturally from the discontinuous discretization.

This paper is organized as follows: In Section 2 we introduce a general class of DG methods. We consider the general framework introduced by Arnold et al. [3], which includes the most relevant DG methods: the interior penalty (IP) [2], the local discontinuous Galerkin (LDG) for purely elliptic problems [10] and a class of non-symmetric methods, called non-symmetric interior penalty Galerkin (NIPG) [6,14]. In Section 3 we prove that, for a particular class of DG methods, the numerical flux coincides with the exact solution at the mesh points for a model problem. In Section 4 we show that a special gradient operator superconverges at the zeros of the Legendre polynomials. Section 5 shows a series of numerical tests that validate our theoretical results. Finally, we end with some concluding remarks in Section 6.

2. Discontinuous Galerkin methods

In this section we describe the formulation of a general class of discontinuous Galerkin methods applied to a two-point boundary value problem

$$\begin{aligned} &-u'' = f \quad \text{in } \Omega = (0,1),\\ &u(0) = u_0, \quad u(1) = u_1. \end{aligned} \tag{1}$$

Based on [3,10], we introduce a new variable $\boldsymbol{q} = u'$. Our model problem can be written as a first-order system of the form

$$\boldsymbol{q} = u' \quad \text{in } \Omega, \tag{2}$$

$$-\boldsymbol{q}' = f \quad \text{in } \Omega, \tag{3}$$

$$u(0) = u_0, \quad u(1) = u_1.$$

Let $\mathscr{T}_h = \{x_i\}_{i=0}^n$ be an arbitrary partition of $\overline{\Omega}$,

$$0 = x_0 < x_1 < \cdots < x_n = 1, \quad I_i = (x_i, x_{i+1}), \quad \overline{\Omega} = \bigcup_{i=0}^{n-1} \overline{I}_i.$$

The weak formulation is obtained by multiplying Eqs. (2) and (3) on each element, I_i, by smooth test functions $\boldsymbol{r}$ and v, respectively. After integrating by parts we obtain the following weak formulation:

$$\int_{I_i} \boldsymbol{q}\boldsymbol{r} = u\boldsymbol{r}(x_{i+1}^-) - u\boldsymbol{r}(x_i^+) - \int_{I_i} u\boldsymbol{r}',$$

$$\int_{I_i} \boldsymbol{q}v' = \boldsymbol{q}v(x_{i+1}^-) - \boldsymbol{q}v(x_i^+) + \int_{I_i} fv.$$

Note that the above equations are well defined for any functions $(u, \boldsymbol{q})$ and $(v, \boldsymbol{r})$ in $\mathscr{V} \times \mathscr{M}$ where

$$\mathscr{V} = \mathscr{M} = \prod_{I_i \in \mathscr{T}_h} H^1(I_i).$$

Next, we seek to approximate the exact solution $(u, \boldsymbol{q})$ with functions $(u_h, \boldsymbol{q}_h)$ in the finite element space $\mathscr{V}_h \times \mathscr{M}_h \subset \mathscr{V} \times \mathscr{M}$, where

$$\mathscr{V}_h = \mathscr{M}_h = \prod_{I_i \in \mathscr{T}_h} \mathscr{P}_p(I_i).$$

The *local* finite element space $\mathscr{P}_p(I_i)$ is the set of polynomials of degree less or equal than p. The finite element solution $(u_h, \boldsymbol{q}_h)$ is defined by using the above weak formulation such that for all $I_i \in \mathscr{T}_h$, and for all $(v, \boldsymbol{r}) \in \mathscr{V}_h \times \mathscr{M}_h$ we have

$$\int_{I_i} \boldsymbol{q}_h\boldsymbol{r} = \hat{\boldsymbol{u}}_h\boldsymbol{r}(x_{i+1}^-) - \hat{\boldsymbol{u}}_h\boldsymbol{r}(x_i^+) - \int_{I_i} u_h\boldsymbol{r}', \tag{4}$$

$$\int_{I_i} \boldsymbol{q}_h v' = \hat{\boldsymbol{q}}_h v(x_{i+1}^-) - \hat{\boldsymbol{q}}_h v(x_i^+) + \int_{I_i} fv, \tag{5}$$

where $\hat{\boldsymbol{u}}_h$ and $\hat{\boldsymbol{q}}_h$ are the so-called numerical fluxes and can be thought of as being approximations to the traces of the function u and $\boldsymbol{q}$, respectively. We assume that these fluxes are local quantities, i.e., they depend only on the traces of the functions $u_h|\{I_{i-1}, I_i\}$, $\boldsymbol{q}_h|\{I_{i-1}, I_i\}$ and/or $u_h'|\{I_{i-1}, I_i\}$, to the mesh point x_i where $\{x_i\} = I_{i-1} \cap I_i$. Moreover, we expect these functions to satisfy some basic properties such as *consistency*, that is, $\hat{\boldsymbol{u}} = u$ and $\hat{\boldsymbol{q}} = u'$, for a smooth function u, typically required in numerical methods for conservation laws. We also expect these functions to satisfy the so-called *conservation* property, which can be formally defined as a numerical flux $\hat{\boldsymbol{\sigma}}(\cdot)$ that is conservative at mesh point x_i, if

$$\hat{\boldsymbol{\sigma}}(x_i^-) = \hat{\boldsymbol{\sigma}}(x_i^+).$$

The choice of the numerical fluxes determines the numerical properties of the method, such as stability, order of convergence, symmetry, etc.; for a detailed theoretical analysis of several DG methods, see the unified framework presented in [3]. In [9], the author performed a theoretical and numerical analysis of the conditioning of some DG methods for a model elliptic problem in 2D. The methods were compared with respect to asymptotic behavior of the condition number, storage cost, rates of convergence, and accuracy of the gradient and the potential.

To define the numerical fluxes we introduced some notation. For any function $v \in \mathscr{V} = \mathscr{M}$, we define the mean, $\{\{v\}\}$, and the jump, $[\![v]\!]$, at mesh point x_i by

$$\{\{v\}\} = \tfrac{1}{2}(v(x_i^-) + v(x_i^+)) \quad \text{and} \quad [\![v]\!] = v(x_i^-) - v(x_i^+).$$

In Table 1, we show the definition of the numerical fluxes for some classical DG methods. Observe that in both the IP [2] and the LDG method [10], the numerical fluxes are conservative and consistent. Finally, for the non-symmetric methods of Baumann and Oden [6] and for the NIPG [14], both fluxes are consistent but only the flux for $\boldsymbol{q}_h$ is conservative.

Table 1
Definition of discontinuous Galerkin methods

Method	$\hat{u}_h$	$\hat{q}_h$
IP [2]	$\{\{u_h\}\}$	$\{\{u_h'\}\} - \frac{\eta_i}{h_i}[\![u_h]\!]$
LDG [10]	$\{\{u_h\}\} + \beta_i \cdot [\![u_h]\!]$	$\{\{q_h\}\} - \beta_i[\![q_h]\!] - \frac{\eta_i}{h_i}[\![u_h]\!]$
Baumann–Oden [6]	$\{\{u_h\}\} + \vec{n}_T \cdot [\![u_h]\!]$	$\{\{u_h'\}\}$
NIPG [14]	$\{\{u_h\}\} + \vec{n}_T \cdot [\![u_h]\!]$	$\{\{u_h'\}\} - \frac{\eta_i}{h_i^{\beta_i}}[\![u_h]\!]$

The definition of the numerical fluxes at the boundary points for the two-point boundary problem (1) are obtained as follows: the flux $\hat{u}_h$ is simply the Dirichlet datum; $\hat{u}_h(0) = u_0$ and $\hat{u}_h(1) = u_1$; and $\hat{q}_h(0)$ is obtained by replacing, in the definition of Table 1, the jump $[\![u_h]\!]$ by $u_0 - u_h(0^+)$ and $\{\{u_h'\}\}$ (or $\{\{q_h\}\} - \beta_i[\![q_h]\!]$) by $u_h'(0^+)$ (or $q_h(0^+)$). For the other boundary points we proceed similarly.

3. Superconvergence at the mesh points

It is well known that for the model problem (1), the approximated solution of the classical finite element method coincides with the exact solution at the mesh points. In this section we show that a similar result also holds for a particular class of DG methods. The major technical problem resides in the lack of continuity of the finite element solution at the grid points. To overcome this difficulty we consider those DG methods with conservative numerical flux $\hat{u}_h$. The conservation property ensures some sort of continuous approximation at these points.

We state the result in the following lemma.

Lemma 1. *Let u_h be the solution of the model problem* (1) *using a DG method with conservative and consistent numerical fluxes $\hat{u}_h$ and $\hat{q}_h$. Then we have*

$$u(x_i) = \hat{u}_h(x_i),$$

where x_i is a mesh point.

Proof. By definition of the numerical flux $\hat{u}_h$ at the boundary points, the result trivially holds for x_0 and x_n. We now consider the case where x_i is an interior grid point. Let $\xi_u = u - u_h$ and $\boldsymbol{\xi}_q = u' - q_h$, then by consistency of both numerical fluxes, ξ_u and $\boldsymbol{\xi}_q$ satisfy the following error equations:

$$\forall \boldsymbol{r} \in \mathscr{M}_h, \quad \int_{I_j} \boldsymbol{\xi}_q \boldsymbol{r} = \hat{\xi}_u \boldsymbol{r}(x_{j+1}^-) - \hat{\xi}_u \boldsymbol{r}(x_j^+) - \int_{I_j} \xi_u \boldsymbol{r}', \tag{6}$$

$$\forall v \in \mathscr{V}_h, \quad \int_{I_j} \boldsymbol{\xi}_q v' = \hat{\boldsymbol{\xi}}_q v(x_{j+1}^-) - \hat{\boldsymbol{\xi}}_q v(x_j^+). \tag{7}$$

Let $v(\cdot) = G_i(\cdot)$ and $\boldsymbol{r}(\cdot) = -G_i'(\cdot)$, where $G_i(x)$ is the function defined by

$$G_i(x) = \begin{cases} x(1-x_i), & 0 \leqslant x \leqslant x_i, \\ (1-x)x_i, & x_i \leqslant x \leqslant 1. \end{cases}$$

Note that $(G_i(\cdot), G_i'(\cdot)) \in \mathscr{V}_h \times \mathscr{M}_h$. After adding the error equation (6) over all j, we obtain the following expression:

$$
\begin{aligned}
-\sum_{j=0}^{n-1}\int_{I_j}\boldsymbol{\xi}_{\boldsymbol{q}}G_i' &= \sum_{j=1}^{n-1}[\![\hat{\xi}_u G_i']\!]_{x_j} + \sum_{j=0}^{n-1}\int_{I_j}\xi_u G_i'' + \hat{\xi}_u G_i'(0) - \hat{\xi}_u G_i'(1),\\
&= \sum_{j=1}^{n-1}[\![\hat{\xi}_u G_i']\!]_{x_j} + \sum_{j=0}^{n-1}\int_{I_j}\xi_u G_i'', \quad \text{since } \hat{\xi}_u(0)=\hat{\xi}_u(1)=0,\\
&= \sum_{j=1}^{n-1}[\![\hat{\xi}_u G_i']\!]_{x_j}, \quad \text{by definition of } G_i,\\
&= \sum_{j=1}^{n-1}\hat{\xi}_u[\![G_i']\!]_{x_j}, \quad \text{by conservation of } \hat{\boldsymbol{u}},\\
&= \hat{\xi}_u[\![G_i']\!]_{x_i}, \quad \text{by definition of } G_i.
\end{aligned}
\tag{8}
$$

Similarly, after adding the error equation (7) over all j, we obtain the following equation:

$$
\begin{aligned}
\sum_{j=0}^{n-1}\int_{I_j}\boldsymbol{\xi}_{\boldsymbol{q}}G_i' &= \sum_{j=1}^{n-1}[\![\hat{\boldsymbol{\xi}}_{\boldsymbol{q}}G_i]\!]_{x_j} + \hat{\boldsymbol{\xi}}_{\boldsymbol{q}}G_i(1) - \hat{\xi}_u G_i(0),\\
&= \sum_{j=1}^{n-1}[\![\hat{\boldsymbol{\xi}}_{\boldsymbol{q}}G_i]\!]_{x_j}, \quad \text{by definition of } G_i,\\
&= \sum_{j=1}^{n-1}\hat{\boldsymbol{\xi}}_{\boldsymbol{q}}[\![G_i]\!]_{x_j}, \quad \text{by conservation of } \hat{\boldsymbol{q}},\\
&= 0, \quad \text{by continuity of } G_i.
\end{aligned}
\tag{9}
$$

The result follows after adding Eqs. (8) and (9), and noticing that $[\![G_i']\!]_{x_i} \neq 0$. □

Remark 1. In both the IP and the LDG method, the fluxes are conservative, thus Lemma 1 holds.

Remark 2. When $\hat{u}_h$ is not conservative Lemma 1 is meaningless as $\hat{u}_h(x_i)$ is a double-valued function. This is the case in both the Baumann–Oden method and the NIPG.

Remark 3. In the LDG method the result holds independently of the choice of the parameters β_i and η_i. Moreover, noticing that the general expression of the numerical flux is

$$\hat{u}_h(x_i) = \left(\tfrac{1}{2}-\beta_i\right)u_h(x_i^+) + \left(\tfrac{1}{2}+\beta_i\right)u_h(x_i^-),$$

we have

$$\beta_i = -0.5 \Rightarrow u(x_i) = \hat{u}_h(x_i) = u_h(x_i^+),$$

$$\beta_i = 0.5 \Rightarrow u(x_i) = \hat{u}_h(x_i) = u_h(x_i^-).$$

Hence, the potential u_h is superconvergent (exact for the model problem) at one of the end-points. This result is analogous to that obtained by Adjerid et al. in [1] for hyperbolic problems. On the other hand this also suggests that the existence of superconvergent points for the potential is much harder to analyze since these will depend on the specific choice of the parameter β_i.

Next, we analyze the local behavior of the auxiliary variable $\boldsymbol{q}_h$. In particular we show that by using conservative numerical fluxes, the auxiliary variable $\boldsymbol{q}_h$ captures the L_2 projection of u' in the local space $\mathscr{P}_{p-1}(I_i)$.

Lemma 2. *Under the assumptions of Lemma* (1) *we have*

$$\forall v \in \mathscr{P}_{p-1}(I_i), \quad \int_{I_i} \boldsymbol{\xi}_{\boldsymbol{q}} v = 0,$$

where $\boldsymbol{\xi}_{\boldsymbol{q}} = u' - \boldsymbol{q}_h$.

Proof. First observe that by letting $\boldsymbol{r} = \chi_{I_i}$, i.e., the characteristic function of I_i, in error equation (6), we have by Lemma (1)

$$\int_{I_i} \boldsymbol{\xi}_{\boldsymbol{q}} \chi_{I_i} = \hat{\xi}_u(x_{i+1}^-) - \hat{\xi}_u(x_i^+) = 0. \tag{10}$$

Now consider the "bubble" functions, $\phi_j(\cdot)$, on the reference element $[-1,1]$, defined by

$$\phi_j(\xi) = \sqrt{\frac{2j-1}{2}} \int_{-1}^{\xi} L_{j-1}(t)\,\mathrm{d}t, \quad \forall j,\ 2 \leqslant j,$$

where $L_j(\cdot)$ is the *Legendre polynomial* of degree j. Without loss of generality we use the same notation for the corresponding "bubble" function in the element I_i, which can be obtained by mapping the reference element to I_i.

Since $\phi_j(\cdot)$ is a polynomial of degree j, we substitute v by ϕ_j in error equation (7). Then, by definition of $\phi_j(\cdot)$ we have

$$\int_{I_i} \boldsymbol{\xi}_{\boldsymbol{q}} \phi_j' = \hat{\boldsymbol{\xi}}_{\boldsymbol{q}} \phi_j(x_{i+1}^-) - \hat{\boldsymbol{\xi}}_{\boldsymbol{q}} \phi_j(x_i^+) = 0, \quad \text{for } j = 2, \ldots, p. \tag{11}$$

Since $\operatorname{span}\{\chi_{I_i}, \phi_2', \ldots, \phi_p'\} = \mathscr{P}_{p-1}(I_i)$, the result follows. □

4. Superconvergence of the gradient at the Gauss–Legendre points

For the class of DG methods presented in Section 2, an approximation of the gradient can be obtained in two ways. The first, u_h', is obtained by applying the classical derivative operator to the computed solution $u_h|_{I_i}$. This operator is completely local. It only requires the degrees of freedom of the cell I_i. The second, $\boldsymbol{q}_h$, is derived from the weak formulation of the discontinuous method, Eq. (2). This is reminiscent of the approximation of the gradient of the classical mixed-finite element method. The operator, although local, requires information from the nearest neighboring cells.

In [8] the author proved that for an unsteady model convection–diffusion problem in 1D using a particular choice of numerical fluxes, the gradient $\boldsymbol{q}_h$ converges in the L_2 norm with an optimal order of $p+1$ when polynomials of degree p are used. This result was then extended to 2D and 3D for elliptic problems on Cartesian meshes [11]. These results motivated us to seek superconvergent points for the gradient $\boldsymbol{q}_h$ instead of the traditional gradient u_h'. Next, we show that the gradient operator $\boldsymbol{q}_h$ superconverges at the zeros of the Legendre polynomial.

Theorem 1 (Superconvergence of $\boldsymbol{q}_h$ at the Gauss–Legendre points). *Under the assumptions of Lemma* 1, *let* $(u_h, \boldsymbol{q}_h) \in \mathscr{P}_p(I_i) \times \mathscr{P}_p(I_i)$ *be the approximated solution of the model problem* (1) *in* I_i. *Then we have*

$$|u'(\eta_l^p) - \boldsymbol{q}_h(\eta_l^p)| = \mathrm{O}(h^{p+1}), \quad l = 1, \ldots, p,$$

where $\{\eta_l^p\}_{1 \leqslant l \leqslant p}$ *are the roots of the Legendre polynomial* $L_p(\cdot)$ *of degree* p *in* I_i.

Proof. We apply the technique presented in [15, p. 35]. Assuming that u is smooth, we expand $\boldsymbol{\xi}_{\boldsymbol{q}}$ in Taylor series around the midpoint of I_i. Then using the Legendre polynomials we have

$$\xi_{\boldsymbol{q}} = \sum_{0}^{p} a_i L_i + \mathrm{O}(h^{p+1}),$$

where $h = x_{i+1} - x_i$. By the orthogonality property of Lemma 2 we have $a_0 = \cdots = a_{p-1} = 0$, so

$$\xi_{\boldsymbol{q}} = a_p L_p + \mathrm{O}(h^{p+1}).$$

Then it is clear that if η_i^p is a zero of the polynomial L_p the result follows. □

5. Numerical experiments

The purpose of this section is to validate the superconvergence result of this paper. We carry out several experiments by numerically solving the model problem (1) using either the IP or the LDG method. For the LDG method, we have set $\beta_i = 0$ and $\eta_i = 1$. With this choice of the parameters β_i and η_i the gradient $\boldsymbol{q}_h$ converges with an order p in the L_2 norm.

In all computations we have used a uniform approximation. In Fig. 1 we show the sets of meshes used in the numerical computations: uniform and non-uniform meshes.

5.1. Arbitrary Dirichlet boundary conditions

The purpose of the following tests is to verify our main result for arbitrary Dirichlet boundary conditions. First, we solve the model problem with zero boundary conditions using the LDG method. The solution is the smooth function $u(x) = 10\sin(5\pi x)$. In Fig. 2 we show the L_∞ error of the gradient $\boldsymbol{q}_h$ at the Gauss–Legendre points, using approximation polynomials of degree $p = 1$–4. In Table 2 we show the numerical order of convergence obtained from linear regression. As predicted by the theory, the gradient $\boldsymbol{q}_h$ converges with an order $p + 1$ when using polynomials of degree p. However, for the local gradient u_h', the Gauss–Legendre points have a convergence of order p only. We have obtained similar results for the IP method.

Next, we solve a model problem with non-zero boundary conditions using the IP method. The solution of this problem is the smooth function $u(x) = 1 + 4.0\sin(x)\mathrm{e}^{\cos(2\pi x)}$. In Fig. 3 we show the L_∞ error of the gradient $\boldsymbol{q}_h$ at the Gauss–Legendre points, using approximation polynomials of degree $p = 1$–4. In Table 3 we show the numerical order of convergence. Again, we obtain an order $p + 1$ for the gradient $\boldsymbol{q}_h$ at the

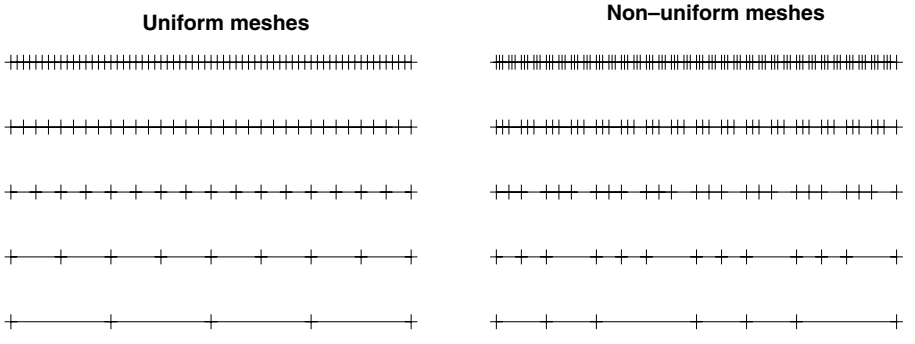


Fig. 1. Sets of meshes: uniform meshes (left), non-uniform meshes (right).

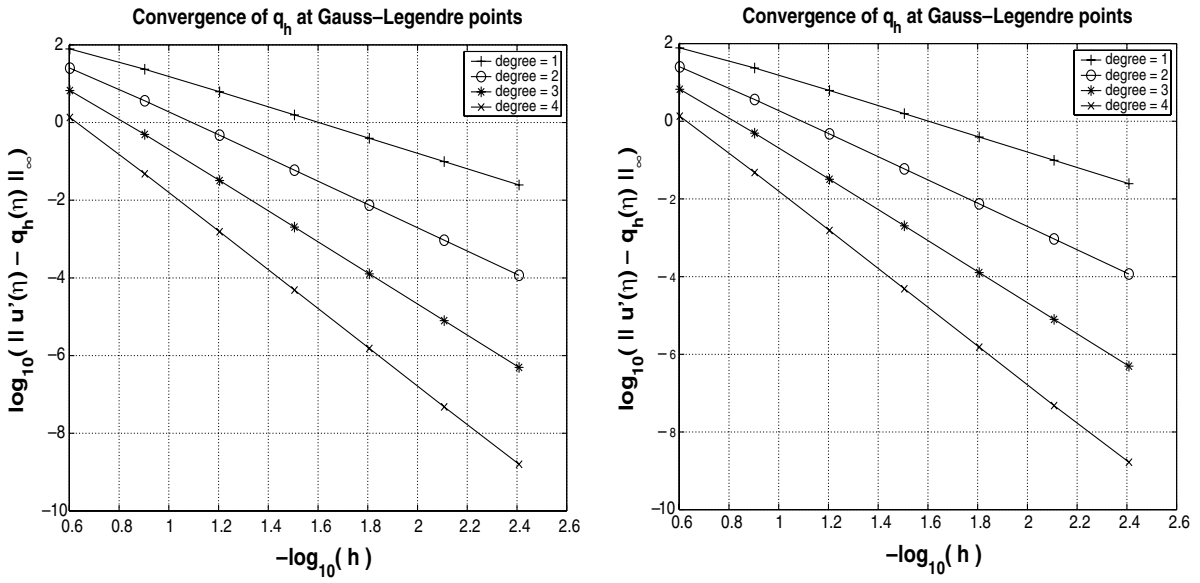


Fig. 2. Convergence in the L_∞ norm of the gradient $\boldsymbol{q}_h$ of the LDG method at the Gauss–Legendre points: uniform meshes (left figure), non-uniform meshes (right figure).

Table 2
Order of convergence at the Gauss–Legendre points for the LDG method

Degree, p	Uniform meshes		Non-uniform meshes	
	$\boldsymbol{q}_h$	u_h'	$\boldsymbol{q}_h$	u_h'
1	1.9533e+00	1.4791e+00	1.9518e+00	1.3951e+00
2	2.9641e+00	1.9466e+00	2.9641e+00	1.9779e+00
3	3.9592e+00	3.2937e+00	3.9592e+00	3.1477e+00
4	4.9590e+00	4.0165e+00	4.9517e+00	3.9752e+00

Gauss–Legendre points but only p for the local gradient u_h'. Analogous results were obtained for the LDG method.

In the following experiments we show that our superconvergence result holds for more general elliptic problems.

5.2. Reaction–diffusion

The purpose of this example is to show that the gradient $\boldsymbol{q}_h$ is superconvergent at the Gauss–Legendre points even in the presence of a reactive term. We consider the following reaction–diffusion problem,

$$-u'' + u = f \quad \text{in } \Omega = (0,1),$$
$$u(0) = u_0 \quad \text{and} \quad u(1) = u_1.$$

The right-hand side term f is chosen such that the exact solution is the smooth function $u(x) = 5e^x - 2\sin(5\pi x)$. In Tables 4 and 5 we present the numerical order of convergence for the IP and LDG methods.

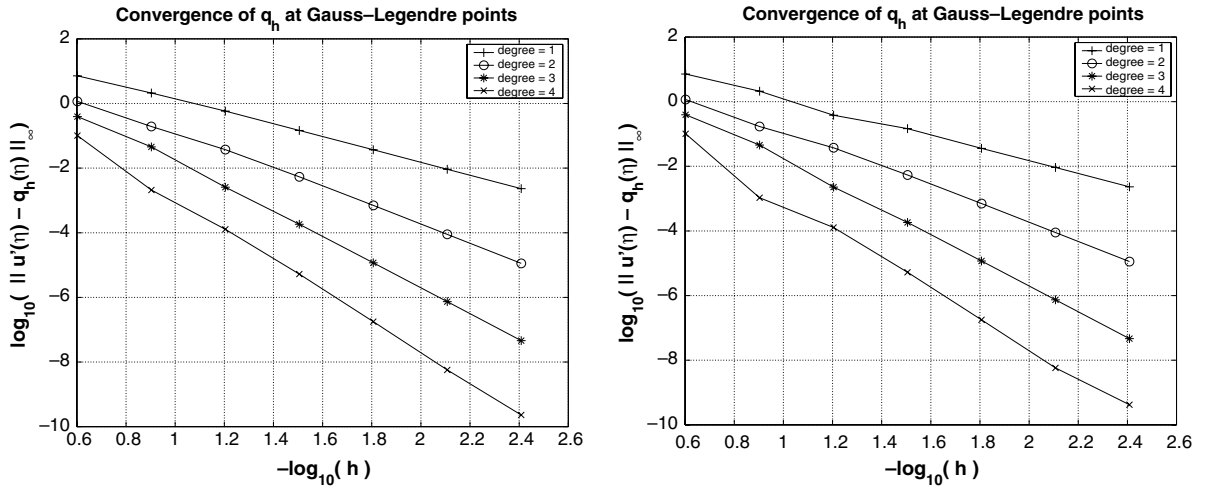


Fig. 3. Convergence in the L_∞ norm of the gradient $\boldsymbol{q}_h$ of the IP method at the Gauss–Legendre points: uniform meshes (left figure), non-uniform meshes (right figure).

Table 3
Order of convergence at the Gauss–Legendre points for the IP method

Degree, p	Uniform meshes		Non-uniform meshes	
	$\boldsymbol{q}_h$	u_h'	$\boldsymbol{q}_h$	u_h'
1	1.9445e+00	8.7983e−01	1.9240e+00	1.1761e+00
2	2.7809e+00	2.2535e+00	2.7676e+00	2.0960e+00
3	3.8813e+00	2.6150e+00	3.8733e+00	2.7864e+00
4	4.7372e+00	3.9893e+00	4.5729e+00	4.2611e+00

Table 4
Convergence at the Gauss–Legendre points for a reactive problem: IP method

Degree, p	Uniform meshes		Non-uniform meshes	
	$\boldsymbol{q}_h$	u_h'	$\boldsymbol{q}_h$	u_h'
1	1.9417e+00	1.7223e+00	1.9384e+00	1.8404e+00
2	2.9538e+00	1.7198e+00	2.9533e+00	1.8637e+00
3	3.9467e+00	3.6530e+00	3.9469e+00	3.4563e+00
4	4.9599e+00	3.5501e+00	4.9579e+00	3.6631e+00

Table 5
Convergence at the Gauss–Legendre points for a reactive problem: LDG method

Degree, p	Uniform meshes		Non-uniform meshes	
	$\boldsymbol{q}_h$	u_h'	$\boldsymbol{q}_h$	u_h'
1	1.9416e+00	1.3024e+00	1.9385e+00	1.4183e+00
2	2.9539e+00	1.9150e+00	2.9536e+00	1.9771e+00
3	3.9470e+00	3.1349e+00	3.9470e+00	3.1602e+00
4	4.9563e+00	4.0088e+00	4.9504e+00	3.9712e+00

Table 6
Convergence at the Gauss–Legendre points for a diffusive problem: IP method

Degree, p	Uniform meshes		Non-uniform meshes	
	$\boldsymbol{q}_h$	u_h'	$\boldsymbol{q}_h$	u_h'
1	1.9439e+00	1.2477e+00	1.9426e+00	1.3372e+00
2	2.9792e+00	2.1034e+00	2.8478e+00	1.9402e+00
3	3.9601e+00	3.0663e+00	3.9721e+00	3.3064e+00
4	4.9490e+00	4.0405e+00	4.6452e+00	3.2394e+00

Table 7
Convergence at the Gauss–Legendre points for a diffusive problem: LDG method

Degree, p	Uniform meshes		Non-uniform meshes	
	$\boldsymbol{q}_h$	u_h'	$\boldsymbol{q}_h$	u_h'
1	1.9484e+00	1.0901e+00	1.9491e+00	1.2343e+00
2	2.9739e+00	2.0998e+00	2.8686e+00	1.9429e+00
3	3.9422e+00	3.0359e+00	3.9632e+00	3.0431e+00
4	4.9407e+00	4.1332e+00	4.6925e+00	3.9375e+00

5.3. Smooth diffusion

In this experiment we consider the purely diffusive problem

$$-(d(x)u')' = f \quad \text{in } \Omega = (0,1),$$
$$u(0) = u_0 \quad \text{and} \quad u(1) = u_1,$$

where the diffusion coefficient $d(x)$ satisfies $0 < d_0 < d(x)$. For this test, the right-hand side was chosen such that the exact solution is the smooth function $u(x) = \sin(x)e^{\sin(5\pi x)}$ and $d(x) = e^{\sin(\pi x)}$. In Tables 6 and 7 we present the numerical order of convergence for the IP and LDG methods.

6. Concluding remarks

In this paper we have performed a numerical and theoretical study of the existence of superconvergent points for DG methods applied to a two-point boundary value problem. We have characterized a class of DG methods for which such points exist. More precisely, we have shown that for a DG method with consistent and conservative numerical fluxes, the gradient operator obtained from the auxiliary variable $\boldsymbol{q}_h$ is superconvergent at the zeros of a Legendre polynomial. However, our numerical experiments have shown that this result does not carry out if we use the traditional gradient operator u_h'. This shows another qualitative difference between these operators.

Our numerical results suggest that the superconvergent behavior of these points holds even for more general problems such as reaction–diffusion problems and diffusion equation with an arbitrary positive diffusivity. The theoretical analysis of this behavior in higher dimensions and for the general case is extremely difficult and is a subject of ongoing research.

This preliminary analysis illustrates new differences among DG methods. The existence of superconvergent points for methods with non-conservative fluxes, such as the NIPG and Baumann–Oden's method, remains an open problem.

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